

Haein Seo

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EDUCATION

Duksung Women's University, Seoul, Republic of Korea

Mar. 2024 – Expected Feb. 2028

B.S., Department of Data Science (*GPA: 3.84/4.5, Major: 3.94/4.5*)

Relevant Coursework:

Introduction to Machine Learning (A+), Introduction to Deep Learning (A+), Big Data Visualization (A+), Data Processing (A+), Big Data Collection & Information Retrieval (A+), Introduction to Big Data (A+)

RESEARCH INTERESTS

- Computer Vision, 3D Scene Understanding, Segmentation, RGB-D Fusion
- Multimodal Learning, VLM, VLA, Document Understanding
- LLM, RAG, Knowledge Graph, Ontology, Persona AI

EXPERIENCE AND TECHNICAL SKILLS

DMKD Lab, Duksung Women's University

Seoul, Korea

Undergraduate Research Assistant

Jan. 2025 – Present

- **Pedestrian-safety vision (first author):** Designed and implemented a street-view object detection pipeline integrated with urban spatial information for pedestrian risk-area analysis, including dataset annotation, data preprocessing, and model fine-tuning; published in a KIPS journal and presented at the KIPS Annual Conference.
- **Explainable medical imaging (first author):** Developed a breast cancer diagnosis framework combining SAM-based lesion segmentation with VLM-based semantic interpretation, along with an XAI-driven clustering and prediction model for severity stratification.
- **Multimodal document & code understanding:** Developed VLM/LLM-based systems for complex document structure understanding (CoDU) and multimodal paper-to-code reconstruction (PRIME-K).
- **Funded projects:** Contributed to an ETRI-commissioned AI image-editing software project and an NRF-funded research project.
- **Research practice:** model implementation, paper reviews, and lab seminars.

Technical Skills

- **Programming & ML Frameworks:** Python, C, PyTorch, TensorFlow, scikit-learn
- **Statistical & Time-Series Analysis:** statsmodels, Pandas, NumPy, Matplotlib
- **Computer Vision & 3D Vision:** Object Detection, Small-Object Detection, Image Segmentation, SAM-based Segmentation, RGB-D Processing, Point Cloud Processing, Point Cloud Completion, 3D Scene Understanding, Visual Representation Learning
- **Multimodal & Vision-Language AI:** Multimodal Learning, Vision-Language Models (VLMs), Vision-Language Reasoning, Document Understanding, Multimodal Document Analysis, LLM, RAG, Explainable AI (XAI)
- **Development & Research Tools:** Git, Docker, Ubuntu/Linux, Hugging Face, CUDA, Overleaf
- **Model Serving & Local Inference:** vLLM, Ollama, OpenClaw

AWARDS AND HONORS

- Gold Prize, SNU FoodTech Commercialization Competition 2026
Seoul National University, Korea, Aug. 2026
- Grand Prize, Department of Data Science Academic Festival 2025
Duksung Women's University, Korea, Nov. 2025

- Best Paper Award, ACK 2025 (Annual Conference of KIPS) 2025
Korea Information Processing Society, Nov. 2025
- Minister's Award, Ministry of Science and ICT (Outreach SW Education Volunteer Group) 2025
Korea, Oct. 2025
- 3rd Place, 2025 Samsung AI Challenge: Visually-rich Document Understanding 2025
Samsung, Korea, Oct. 2025

SCHOLARSHIPS

- KT Digital Talent Scholarship 2026
KT, Mar. 2026 – Aug. 2026
- Academic Excellence Scholarship 2024
1st in Department (1st Semester 2024 / GPA: 4.25) – Duksung Women's University
Top 10% (2nd Semester 2024 / GPA: 4.17) – Duksung Women's University

LEADERSHIP & ACTIVITIES

- AI Team, KT Social Problem-Solving Project 2026
KT
- Team Lead, Outreach SW Education Volunteer Group 2025
Code Club, Korea
- Administrative Dept., Hanbok Club 2025
Duksung Women's University, Korea
- Cooperation Team, University Career Supporters 2024–2025
Career Development Center, Duksung Women's University, Korea
- SW Education Mentor, Shinsegae I&C & KOFAC Coding Programs 2024
Shinsegae I&C / Korea Foundation for the Advancement of Science and Creativity

PUBLICATIONS

Journal Articles

- 1) **Haein Seo**, Eun-Chong Lee, and Jehyeok Rew, “Pedestrian Risk Area Analysis Using Street View-based Object Detection and Urban Spatial Information,” *The Transactions of the Korea Information Processing Society*, vol. 15, no. 8, pp. 681–695, 2026.
- 2) **Haein Seo** and Jehyeok Rew, “An Explainable Breast Cancer Diagnosis Framework using SAM-based Lesion Segmentation and VLM-based Semantic Interpretation,” *Journal of Platform Technology*, vol. 14, no. 3, pp. 59–70, 2026. doi:10.23023/JPT.2026.14.3.059

Conference Proceedings

- 1) **H.-I. Seo**, E.-C. Lee, and J. Rew, “Pedestrian safety monitoring method using street view object detection and urban spatial information analysis,” in *Proc. Annu. Conf. KIPS*, 2025, pp. 520–521.
- 2) **H.-I. Seo**, S.-M. Seo, Y.-J. Lee, S.-H. Kim, and J. Moon, “Hybrid clustering-prediction model and XAI analysis for severity stratification in malignant breast cancer patients,” in *Proc. ICT Platform Soc. Fall Conf.*, Seongnam, South Korea, 2025, pp. 66–70.
- 3) K. Kim, M. Lee, **H. Seo**, and J. Rew, “CoDU: Multi-Stage Approach for Complex Document Structure Understanding,” in *Proc. Korea Database Conf. (KDBC)*, Gyeongju, Republic of Korea, Nov. 2025, pp. 167–170.
- 4) **H. Seo** and J. Rew, “PRIME-K: A Multimodal Paper-to-Code Reconstruction Framework for Keystroke Dynamics Research,” in *Proc. Korea Computer Congress (KCC)*, Republic of Korea, 2026, pp. 522–524.

CERTIFICATE AND LICENSE

- Google Analytics Certification, Google Sep. 2024